

Clinical Outcomes After Unilateral Adrenalectomy for Primary Aldosteronism

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IMPORTANCE In addition to biochemical cure, clinical benefits after surgery for primary aldosteronism depend on the magnitude of decrease in blood pressure (BP) and use of antihypertensive medications with a subsequent decreased risk of cardiovascular and/or cerebrovascular morbidity and drug-induced adverse effects.

OBJECTIVE To evaluate the change in BP and use of antihypertensive medications within an international cohort of patients who recently underwent surgery for primary aldosteronism.

DESIGN, SETTING, AND PARTICIPANTS A cohort study was conducted across 16 referral medical centers in Europe, the United States, Canada, and Australia. Patients who underwent unilateral adrenalectomy for primary aldosteronism between January 2010 and December 2016 were included. Data analysis was performed from August 2017 to June 2018. Unilateral disease was confirmed using computed tomography, magnetic resonance imaging, and/or adrenal venous sampling. Patients with missing or incomplete preoperative or follow-up data regarding BP or corresponding number of antihypertensive medications were excluded.

MAIN OUTCOMES AND MEASURES Clinical success was defined based on postoperative BP and number of antihypertensive medications. Cure was defined as normotension without antihypertensive medications, and clear improvement as normotension with lower or equal use of antihypertensive medications. In patients with preoperative normotensivity, improvement was defined as postoperative normotension with lower antihypertensive use. All other patients were stratified as no clear success because the benefits of surgery were less obvious, mainly owing to postoperative, persistent hypertension. Clinical outcomes were assessed at follow-up closest to 6 months after surgery.

RESULTS On the basis of inclusion and exclusion criteria, a total of 435 patients (84.6%) from a cohort of 514 patients who underwent unilateral adrenalectomy were eligible. Of these patients, 186 (42.3%) were women; mean (SD) age at the time of surgery was 50.7 (11.4) years. Cure was achieved in 118 patients (27.1%), clear improvement in 135 (31.0%), and no clear success in 182 (41.8%). In the subgroup classified as no clear success, 166 patients (91.2%) had postoperative hypertension. However, within this subgroup, the mean (SD) systolic and diastolic BP decreased significantly by 9 (22) mm Hg ($P < .001$) and 3 (15) mm Hg ($P = .04$), respectively. Also, the number of antihypertensive medications used decreased from 3 (range, 0-7) to 2 (range, 0-6) ($P < .001$). Moreover, in 75 of 182 patients (41.2%) within this subgroup, the decrease in systolic BP was 10 mm Hg or greater.

CONCLUSIONS AND RELEVANCE In this study, for most patients, adrenalectomy was associated with a postoperative normotensive state and reduction of antihypertensive medications. Furthermore, a significant proportion of patients with postoperative, persistent hypertension may benefit from adrenalectomy given the observed clinically relevant and significant reduction of BP and antihypertensive medications.

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Primary aldosteronism (PA) is the most common form of secondary hypertension. The disease is characterized by inappropriate endogenous production of the mineralocorticoid aldosterone by one or both of the adrenal glands.^{1,2} Prevalence is estimated at 5% in the general hypertensive population and even higher in populations with severe or resistant hypertension.³⁻⁵ Due to aldosteronism itself and subsequent hypertension, PA leads to long-term fibrosis and remodeling in critical organs resulting in an increased risk of cardiovascular, cerebrovascular, and renal morbidity and mortality.⁶⁻⁹ Therefore, PA could be considered a serious health issue.^{9,10}

In most cases, PA is accounted for by either an aldosterone-producing adenoma, which is generally treated with adrenalectomy, or bilateral adrenal hyperplasia, which is treated with mineralocorticoid receptor antagonists.¹⁰ Adequate treatment of PA leads to significant reduction of morbidity and mortality through cure or improvement of aldosteronism and hypertension.^{11,12} Biochemical cure (ie, normalization of plasma aldosterone levels) is achieved in almost all patients following adrenalectomy (96%-100%).¹³ However, results on clinical cure (ie, postoperative normotensive state without the use of antihypertensive medications) vary extensively across studies (22%-84%).¹³⁻¹⁵

Systematic reviews and meta-analyses have indicated clinical cure on pooled data in 42%, 50%, and 52% of patients.¹³⁻¹⁵ These reviews include numerous studies presenting clinical outcomes after adrenalectomy published over the last few decades. However, most included studies were single center with small populations. In studies with larger populations, the cohort was frequently spanning multiple decades with the potential introduction of bias because of improvement in diagnosis, workup, and treatment of PA with updated guidelines as well as innovations in diagnostic modalities and surgical techniques over time. Moreover, the worldwide increase in hypertension over the last few decades could influence the hypertension-related outcomes after surgery owing to the increase in hypertension not associated with PA (ie, background or essential).¹⁶ Furthermore, these studies were mainly focused on presenting proportions of patients with clinical cure or improvement and identifying possible prognostic factors instead of describing the decrease of BP and number of antihypertensive medications, which is important for daily clinical practice.^{17,18}

Because clinical benefits of surgery mostly depend on the magnitude of BP decrease rather than crossing the BP threshold that currently defines hypertension, we hypothesized that precise presentation of a decrease in BP and use of antihypertensive medications after adrenalectomy leads to better understanding of the benefits of surgery in PA.¹⁹ Therefore, we set out to investigate and precisely display the association of adrenalectomy with BP and the need for antihypertensive medications in a large, international cohort of patients who underwent adrenalectomy for PA between 2010 and 2016. In addition, we hypothesized that patients in whom the benefits of surgery are less obvious, for instance, due to persistent hypertension after adrenalectomy, could benefit from surgery, especially because every 10-mm Hg reduction in systolic BP

Key Points

Question Is adrenalectomy associated with reduction of blood pressure and need for antihypertensive medications in patients with primary aldosteronism?

Findings In this international cohort study of 435 surgical patients between 2010 and 2016, 27.1% of patients achieved normotension without requiring antihypertensive medications and 31.0% achieved normotension requiring less than or equal to the number of their preoperative antihypertensive medications. Moreover, patients with postoperative persistent hypertension might have benefitted from surgery given the observed significant reduction of blood pressure and number of medications within this subgroup.

Meaning Most patients may benefit from adrenalectomy owing to a decrease in blood pressure and need for antihypertensive medications.

results in significant lowering of cardiovascular morbidity and mortality.¹⁹ Therefore, we chose to precisely illustrate the outcome of surgery within this specific subgroup as well.

Methods

Patients

We performed an international, retrospective cohort study across 16 referral medical centers in the United States, Europe, Canada, and Australia (Table 1). All patients who underwent unilateral adrenalectomy between January 2010 and December 2016 for aldosterone-producing adenoma, proven by computed tomographic, and/or magnetic resonance imaging, and/or adrenal venous sampling were included. Patients with missing or incomplete preoperative or follow-up data regarding systolic BP (SBP), diastolic BP (DBP), or corresponding number of antihypertensive medications were excluded. Data collection was performed separately within each center with the use of a standardized data-entry manual. Patient demographics, disease characteristics, laboratory data (eg, measurements of aldosterone-to-renin ratio and potentially confirmatory tests), results of computed tomography, magnetic resonance imaging, adrenal venous sampling operative characteristics, pathologic testing diagnosis, and timing of follow-up were collected. To compare laboratory data between centers, measurements were classified as elevated or suppressed when values were above or below the local reference ranges. Institutional review board approval was obtained in all participating centers; excluded the need for informed consent was waived owing to the retrospective nature of the study. Patients had not received financial compensation.

Outcomes and Definitions

The primary outcomes of this study were the preoperative to postoperative change in SBP and DBP (millimeters of mercury) with subsequent change in the use of antihypertensive medications. If multiple preoperative or postoperative BP measurements were performed during therapy with the same antihypertensive medications, the mean SBP and DBP were

Table 1. Baseline Characteristics of 435 Patients

Variable ^a	No. (%)
Age at surgery, mean (SD), y	50.7 (11.4)
Women	186 (42.8)
Duration of hypertension (n = 366), y ^b	
Median (range)	9 (0-42)
BMI (n = 402)	
Mean (SD)	29.7 (6.0)
No. of antihypertensives, median (range) ^a	3 (0-8)
Hypokalemia, No. (n = 429)	317 (73.9)
History of cardiovascular events, No. (n = 431)	62 (14.4)
Diabetes (n = 432)	57 (13.2)
Current smoker (n = 417)	48 (11.5)
Hypercholesterolemia (n = 430)	114 (26.5)
Family history of hypertension (n = 338)	173 (51.2)
Preoperative BP, mean (SD), mm Hg	
Systolic	150 (20)
Diastolic	90 (13)
JNC/ESH hypertension grade based on BP with medication ^c	
0	111 (25.5)
1	180 (41.4)
2	105 (24.1)
3	39 (9.0)
Elevated aldosterone level (n = 408)	225 (55.1)
Suppressed renin level/activity (n = 370)	245 (66.2)
ARR indicating PA (n = 361)	341 (94.5)
Elevated creatinine level (n = 392)	71 (18.1)
CT performed (n = 432)	378 (87.1)
AVS performed (n = 434)	278 (64.1)
MRI performed (n = 434)	72 (16.6)
Confirmatory test performed	143 (32.9)
Oral salt loading	18 (4.1)
Saline infusion test	118 (27.1)
Fludrocortisone suppression test	3 (<1)
Captopril challenge	1 (<1)
Fludrocortisone dexamethasone suppression test	1 (<1)
Post-low-dose dexamethasone suppression-saline infusion test	1 (<1)
Surgical procedure	
EPRA	171 (39.3)
ELRA	65 (14.9)
LTA	198 (45.5)
Open	1 (<1)
Robot assisted	17 (3.9)
Conversion	2 (<1)
Tumor laterality	
Left	260 (60)
Right	175 (40.2)
Histologic diagnosis	
Adenoma	362 (83.2)
Hyperplasia	58 (13.3)
Adenoma/hyperplasia	13 (3.0)
Missing	1 (<1)
Hospital stay, median (range), d ^b	1 (0-70)

(continued)

Table 1. Baseline Characteristics of 435 Patients (continued)

Variable ^a	No. (%)
Follow-up after surgery, mo	
<1	101 (23.2)
1-<3	39 (9.0)
3-9	278 (63.9)
>9-12	4 (<1)
>12-18	13 (3.0)

Abbreviations: ARR, aldosterone-to-renin-ratio; AVS, adrenal venous sampling; BMI, body mass index (calculated as weight in kilograms divided by height in meters squared); BP, blood pressure; CT, computed tomography; ELRA, endoscopic lateral retroperitoneal adrenalectomy; EPRA, endoscopic posterior retroperitoneal adrenalectomy; ESH, European Society of Hypertension; JNC, Joint National Commission; LTA, laparoscopic transabdominal adrenalectomy; MRI, magnetic resonance imaging; PA, primary aldosteronism; SBP, systolic blood pressure.

^a Numbers in parentheses indicate sample sizes different from the total population.

^b Values not normally distributed given as medians (range).

^c Grade 0, SBP less than 140 mm Hg and diastolic BP (DBP) less than 90 mm Hg; grade 1, SBP 140 to 159 mm Hg and/or DBP 90 to 99 mm Hg; grade 2, SBP 160 to 179 mm Hg and/or DBP 100 to 109 mm Hg; and grade 3, SBP 180 mm Hg or higher or DBP 110 mm Hg or higher.^{20,21}

calculated. In general, office BP measurements were performed during outpatient visits; however, 24-hour ambulatory BP measurements were preferred. The number, names, and dosages of different antihypertensive medications used at the time of BP measurements were collected. When medications were withheld for diagnostic testing, such as the aldosterone-to-renin ratio or a confirmatory test, SBP and DBP with corresponding medications before discontinuation were used. The number of antihypertensive medications was defined as the number of different antihypertensive medication categories used (eg, calcium channel blockers, β -blockers). If data were sufficient, the defined daily dose (DDD), based on the World Health Organization Anatomical Therapeutic Chemical DDD Index 2017,²² and the number of pills taken by the patient each day were calculated.

Hypertension grade, as established by the European Society of Hypertension and Joint National Commission, was based on BP during medication therapy.^{20,21} Grade 0 was defined as SBP less than 140 mm Hg and DBP less than 90 mm Hg; grade 1, SBP 140 to 159 mm Hg and/or DBP 90 to 99 mm Hg; grade 2, SBP 160 to 179 mm Hg and/or DBP 100 to 109 mm Hg; and grade 3, SBP 180 mm Hg or higher or DBP 110 mm Hg or higher. Clinical success was stratified as cure, clear improvement, or no clear success based on postoperative SBP and DBP and number of antihypertensive medications. Cure was defined as a postoperative normotensive patient (ie, SBP <140 and DBP <90 mm Hg) without the need for antihypertensive medication.^{20,21} Clear improvement was defined as postoperative patients with normal BP receiving a lower or equal number of antihypertensive medications. For a preoperative patient with normal BP, a decrease in the number of antihypertensive medications was required. All other patients were stratified as no clear success because the possible benefits of surgery within this subgroup were less obvious, mainly owing to persistent hypertension after surgery. We also

Table 2. Distribution of Included Patients per Center and Period of Follow-up

Medical Center	No. (%)		Period of Follow-up After Surgery, mo				
	2010-2016 ^a						
	Received Surgery	Eligible	<1	1-<3	3-9	>9-12	>12-18
University of California San Francisco Medical Center	82	69 (84.1)	49 (71.0)	5 (7.2)	15 (22)	0	0
Northwestern Memorial Hospital	51	43 (84.3)	19 (44.2)	4 (9.3)	19 (44.2)	1 (2.3)	0
Royal North Shore Hospital	51	39 (76.5)	0	0	39 (100)	0	0
Weill Cornell Medical Center	47	40 (85.1)	2 (5.0)	0	37 (92.5)	0	1 (2.5)
Columbia University Medical Center	46	40 (87.0)	3 (7.5)	1 (2.5)	36 (90.0)	0	0
University Health Network Toronto	44	32 (72.7)	2 (6.3)	8 (25.0)	19 (59.4)	0	3 (9.4)
University Medical Center Groningen	41	36 (87.8)	0	2 (5.6)	28 (77.8)	2 (5.6)	4 (11.1)
University Medical Center Utrecht	40	37 (92.5)	2 (5.4)	5 (13.5)	28 (75.7)	1 (2.7)	1 (2.7)
University of Chicago Medical Center	26	25 (96.2)	15 (60.0)	0	10 (40.0)	0	0
MD Anderson Cancer Center	19	19 (100)	6 (31.6)	6 (31.6)	6 (31.6)	0	1 (5)
Instituto de Semeiotica Cirurgica Roma	16	9 (56.3)	0	0	9 (100)	0	0
Academic Medical Center Amsterdam	15	11 (73.3)	0	0	11 (100)	0	0
Boston Medical Center	12	12 (100)	1 (8.3)	3 (25.0)	8 (66.7)	0	0
Montreal General Hospital-McGill University Health Center	10	10 (100)	1 (10.0)	1 (10.0)	8 (80.0)	0	0
VU University Medical Center	8	8 (100)	0	4 (50.0)	1 (12.5)	0	3 (37.5)
University Medical Center Maastricht	6	5 (83.3)	1 (20.0)	0	4 (80.0)	0	0
Total	514	435 (84.6)	101 (23.2)	39 (9.0)	278 (63.9)	4 (1.0)	13 (3.0)

^a The median number of patients who underwent surgery and were eligible per medical center were 33 (range, 6-82) and 28.5 (range, 5-69), respectively.

stratified categories based on the magnitude of SBP decrease (ie, <10, 10-19, 20-29, 30-39, 40-49, and ≥50 mm Hg). For this stratification, all patients with an increase in the number of antihypertensive medications were excluded to minimize the possible effect of increased medication on decrease in SBP. The goal was to assess the primary outcomes at follow-up closest to 6 months after adrenalectomy (range, 3-9 months).

Mainly because of geographic distances, multiple medical centers were not able to complete this 6 months of follow-up (range, 3-6 months). To prevent a high percentage of patients being lost to follow-up, we included patients who underwent follow-up during other periods in the analysis.

Statistical Analysis

Data analysis was performed between August 2017 and June 2018. Normally and not normally distributed continuous data are shown as mean (SD) and median (range). The McNemar test was used for paired nominal data, the paired-sample *t* test for paired, normally distributed continuous data, and the Wilcoxon signed ranked test for paired, not normally distributed continuous data. To compare continuous variables between groups, the Mann-Whitney test (2 groups) or Kruskal-Wallis test (>2 groups) was used for not normally distributed data and 1-way analysis of variance for normally distributed data. The χ^2 test and Fisher exact test were used to analyze group differences for categorical variables, with *P* < .05 considered significant. Statistical analysis was performed using SPSS, version 23.0 (SPSS Institute).

Results

In total, 514 patients underwent unilateral adrenalectomy for PA between 2010 and 2016. Based on inclusion and exclusion criteria, 435 patients (84.6%) were eligible for further analysis. The primary reason for exclusion was inadequate preoperative and/or postoperative data regarding BP and number of antihypertensives. The cohort included 186 women (42.3%), and the mean (SD) age at the time of surgery was 50.7 (11.4) years. Hypokalemia was present in 74% of patients and 341 of 361 patients with data available (94.5%) had an aldosterone-to-renin ratio indicating PA. Computed tomographic, adrenal venous sampling, and magnetic resonance imaging were performed in 88%, 64%, and 17% of patients, respectively. Further baseline characteristics are reported in **Table 2**. Distribution of patients who underwent operations and the period of follow-up per medical center are presented in **Table 1**.

Overall Association Between Surgery, BP, and Antihypertensives

In the entire cohort, the preoperative mean (SD) SBP and DBP were 150 (20) and 90 (13) mm Hg. Grade 1 hypertension was most frequent (180 [41.4%]). The preoperative median number of antihypertensive medications used was 3 (range, 0-8); defined daily doses, 3.7 (range, 0.0-25.3); and number of pills taken each day, 3 (range, 0-10). After surgery, the mean (SD) SBP decreased to 133 (16) mm Hg, and the DBP dropped to 83 (10) mm Hg, resulting in reductions of 17 (21) mm Hg (10.3% [13.2%]) and 7 (14 mm Hg) (7.0% [14.5%]), respectively (all

$P < .001$). Also, reductions were noted in the median number of antihypertensives (from 3 [range, 0-8] to 1 [range, 0-6]; 60.0%), DDDs (from 3.7 [range, 0.0-25.3] to 1.0 [range, 0.0-11.7]; 72.7%), and number of pills (from 3 [range, 0-3] to 1 [range, 0-9]; 66.7%) (all $P < .001$). A total of 269 patients (61.8%) had grade 0 hypertension and therefore were normotensive after surgery (including patients using and not using antihypertensive medications).

Cure, Clear Improvement, and No Clear Success

Cure was achieved in 118 (27.1%) patients, clear improvement in 135 (31.0%) patients, and no clear success in 182 (41.8%) patients (Table 3). No clear differences in the proportions of patients with cure, clear improvement, and no clear success were shown between the 5 different periods of follow-up ($P = .28$). Within the group stratified as cure, significant decreases were seen in the mean SBP (21 [18] mm Hg; 13.4% [11.2%]) and DBP (9 [11] mm Hg; 9.6% [12.9%]) (both $P < .001$). As per definition, within this subgroup, all antihypertensive medications were stopped. Significant decreases were also seen within the group stratified as clear improvement in the mean SBP (25 [18] mm Hg; 15.6% [10.1%]) and DBP (12 [12] mm Hg; 11.9% [12.6%]) (both $P < .001$). Reductions were noted in the median number of antihypertensives (50.1%), DDDs (48.3%), and number of pills (50.0%) (all $P < .001$). In the group stratified as no clear success, the mean SBP decreased significantly by 9 (22) mm Hg (4.3% [14.0%]; $P < .001$), and DBP was decreased significantly by 3 (15) mm Hg (1.5% [15.1%]; $P = .04$). Significant reductions were noted in the median number of antihypertensives (50.0%), DDDs (53.1%), and number of pills (50.0%) (all $P < .001$). Pairwise comparison between the 3 groups showed similar magnitude of decrease in SBP and DBP between cure and clear improvement. Furthermore, the decrease in DDD was comparable between the 3 groups (median, 2.0; range, -4.7 to 24.3; $P = .52$), resulting from the significant lower preoperative DDD within patients with cure (median, 2.2; range, 0.0 to 15.7) compared with clear improvement (median, 1.9; range, -4.7 to 24.3; $P < .001$) and no clear success (median, 2.0; range, -4.3 to 20.7; $P < .001$). Geographic stratification of the rates of clinical success and other BP-related outcomes for the United States, Europe, Canada, and Australia is presented in the eTable in the Supplement.

Subanalysis of patients classified as having no clear success, in which the benefits of surgery were less obvious, is reported in Table 4. This group consisted of 166 (91.2%) patients with postoperative hypertension and 16 (8.8%) patients who achieved normotension after surgery but showed an increase in the number of antihypertensives or no decrease in the number of antihypertensives in cases with preoperative normotension. Within the group with postoperative hypertension, 26 patients (15.7%) had normotension before surgery and showed an increase in mean SBP (17 [14] mm Hg) and DBP (9 [9] mm Hg) (both $P < .001$) after surgery. However, the number of antihypertensives (66.7%), DDDs (82.6%), and number of pills (66.7%) decreased significantly (all $P < .001$). The other 140 (84.3%) patients had hypertension before and after surgery, however, showed a significant reduction of mean SBP by 13 (21) mm Hg ($P < .001$) and DBP by 4 (15) mm Hg ($P = .001$).

Also, a significant decrease was shown in the number of antihypertensives (50.0%), DDDs (57.7%), and number of pills (50.0%) (all $P < .001$).

Within the 16 patients who had normotension after surgery, 9 (56.3%) were normotensive before and after surgery. They showed no significant change in BP or number of antihypertensive medications. The other 7 (43.8%) patients had hypertension before the operation and, although mean SBP (29 [15] mm Hg; $P = .02$) and DBP (18 [14] mm Hg; $P = .03$) were decreased significantly in these patients, they were classified as no success owing to an increase in the number of antihypertensives.

Magnitude of Change in SBP

Fourteen of 435 patients (3.2%) showed an increase in the number of antihypertensives after surgery; therefore, a possible decrease in SBP could be due to medication. The remaining 421 patients (96.8%) showed a decreased or equal number of antihypertensives after surgery. In the total population, 76 patients (17.5%) had an increase in SBP. However, 12 of these patients (15.8%) continued to have normotension without antihypertensive medication (cure) and 8 patients (10.5%) had normotension with the decreased or equal number of antihypertensives (clear improvement). Seventy-one patients (16.3%) showed a decrease in SBP between 0 and 9 mm Hg; 87 patients (20.0%), a decrease between 10 and 19 mm Hg; 84 (19.3%) patients, a decrease between 20 and 29 mm Hg; 51 (11.7%) patients, a decrease between 30 and 39 mm Hg; 24 (5.5%) patients, a decrease between 40 and 49 mm Hg; and 28 (6.6%) patients, a decrease of 50 mm Hg or more (Table 5).

Within the subgroups stratified as cure and clear improvement, a decrease in SBP between 20 and 29 mm Hg (29 [24.6%]) and 10 and 19 mm Hg (35 [25.9%]) were most frequent. Within the subgroup stratified as no clear success, an increase in SBP was most frequent (56 [30.8%]). However, 75 (41.2%) patients within the no clear success subgroup had a decrease in SBP of 10 mm Hg or more.

Discussion

Normalization of hyperaldosteronism after adrenalectomy for PA, which is shown in most cases, does not always lead to normalization of the BP. Therefore, the assessment of clinical success (ie, decrease in BP and/or number of antihypertensive medications) after adrenalectomy is an important indicator for surgical outcome. This study describes the association of adrenalectomy with blood pressure and use of antihypertensive medications within a global cohort of patients undergoing surgery for PA between 2010 and 2016. Although most patients (58.2%) showed cure or clear improvement of hypertension after surgery, results also suggested beneficial clinical outcomes of surgery (ie, reduction of BP and/or number of antihypertensive medications) in a large proportion of patients with persistent hypertension after surgery. This is best highlighted within the 182 (41.8%) patients stratified as no clear success in whom the benefits of surgery were less obvious. In addition, in 41.2% (15% of the total population) of patients

Table 3. Clinical Success: Cure, Clear Improvement, and No Clear Success

Variable	Total Cohort (n = 435)	Clinical Success			P Value for Comparison Between Groups			
		Cure (n = 118 [27.1%])	Clear Improvement (n = 135 [31.0%])	No Clear Success (n = 182 [41.8%])	Overall	Pairwise		
						Cure vs Clear Improvement	Cure vs No Clear Success	Clear Improvement vs No Clear Success
Preoperative								
SBP, mean (SD), mm Hg	150 (20)	143 (17)	150 (17)	155 (22)	<.001	.007	<.001	.07
DBP, mean (SD), mm Hg	90 (13)	88 (11)	90 (12)	91 (15)	.07	.56	.07	>.99
Preoperative hypertension grade, No. (%) ^a								
0	111 (25.5)	49 (41.5)	27 (20.0)	35 (19.2)	<.001	<.001	<.001	.03
1	180 (41.4)	41 (34.7)	70 (51.9)	69 (37.9)				
2	105 (24.1)	24 (20.3)	27 (20.0)	54 (29.7)				
3	39 (9.0)	4 (3.4)	11 (8.1)	24 (13.2)				
No. of antihypertensives/d, median (range) ^b	3 (0 to 8)	2 (0 to 6)	3 (1 to 8)	3 (0 to 7)	<.001	<.001	<.001	.27
DDD (n = 405) ^b								
Median (range)	3.7 (0.0 to 25.3)	2.2 (0.0 to 15.7)	4.1 (0.5 to 25.3)	4.3 (0.0 to 22.3)	<.001	<.001	<.001	.86
No. of pills/d (n = 407) ^b								
Median (range)	3 (0 to 10)	2 (0 to 7)	4 (1 to 10)	4 (0 to 9)	<.001	<.001	<.001	.22
Postoperative								
SBP, mean (SD), mm Hg	133 (16)	122 (9)	126 (9)	147 (15)	<.001	.12	<.001	<.001
DBP, mean (SD), mm Hg	83 (10)	78 (7)	78 (7)	89 (11)	<.001	>.99	<.001	<.001
Preoperative hypertension grade, No. (%) ^a								
0	269 (61.8)	118 (100)	135 (100)	16 (8.8)	<.001	>.99	<.001	<.001
1	117 (26.9)	0	0	117 (64.3)				
2	41 (9.4)	0	0	41 (22.5)				
3	8 (1.8)	0	0	8 (4.4)				
No. of antihypertensives/d, median (range) ^b	1 (0 to 6)	0	2 (1 to 5)	2 (0 to 6)	<.001	<.001	<.001	.30
DDD (n = 402) ^b								
Median (range)	1.0 (0.0 to 11.7)	0	2.0 (0.1 to 11.7)	2.0 (0.0 to 9.8)	<.001	<.001	<.001	.25
No. of pills/d (n = 407) ^b								
Median (range)	1 (0 to 9)	0	2 (0 to 8)	2 (0 to 9)	<.001	<.001	<.001	.38
Preoperative-Postoperative Change^c								
SBP, mean (SD), mm Hg	17 (21)	21 (18)	25 (18)	9 (22)	<.001	.28	<.001	<.001
DBP, mean (SD), mm Hg	7 (14)	9 (11)	12 (12)	3 (15)	<.001	.49	<.001	<.001
SBP, mean (SD), %	10.3 (13.2)	13.4 (11.2)	15.6 (10.1)	4.3 (14.0)	<.001	.45	<.001	<.001
DBP, mean (SD), %	7.0 (14.5)	9.6 (12.9)	11.9 (12.6)	1.5 (15.1)	<.001	.55	<.001	<.001
No. of antihypertensives/d ^b								
Median (range)	2 (−3 to 6)	2 (0 to 6)	1 (0 to 5)	1 (−3 to 5)	<.001	<.001	<.001	.58
Median (range), %	60 (−100 to 100)	100 (0 to 100)	50 (0 to 80)	50 (−100 to 100)	<.001	<.001	<.001	.68
DDD (n = 382) ^b								
Median (range)	2.0 (−4.7 to 24.3)	2.2 (0.0 to 15.7)	1.9 (−4.7 to 24.3)	2.0 (−4.3 to 20.7)	.52	.28	.36	.81
Median (range), %	73 (−400 to 100)	100 (0 to 100)	48 (−275 to 99)	53 (−400 to 100)	<.001	<.001	<.001	.05
No. of pills/d (n = 388) ^b								
Median (range)	2 (−3 to 8)	2 (0 to 7)	2 (−2 to 8)	2 (−3 to 6)	.02	.11	.005	.21
Median (range), %	67 (−200 to 100)	100 (0 to 100)	50 (−200 to 100)	50 (−100 to 100)	<.001	<.001	<.001	.85

Abbreviations: CIMP, clear improvement; DBP, diastolic blood pressure; DDD, defined daily dose; ESH, European Society of Hypertension; JNC, Joint National Commission; SBP, systolic blood pressure.

^a Grade 0, SBP less than 140 mm Hg and DBP less than 90 mm Hg; grade 1, SBP 140 to 159 mm Hg and/or DBP 90 to 99 mm Hg; grade 2, SBP 160 to 179 mm Hg and/or DBP 100 to 109 mm Hg; and grade 3, SBP 180 mm Hg or higher or

DBP 110 mm Hg or higher.^{20,21}

^b Values not normally distributed given as medians (range).

^c Within all outcome variables (also within cure, clear improvement, and no clear success) the preoperative-postoperative Δ values showed significant decrease ($P < .05$).

Table 4. Subgroup Analysis of Patients Stratified as No Success

Variable	No Clear Success (n = 182)	Postoperative Hypertension (n = 166 [91.2%])	Postoperative Normotension (n = 16 [8.8%])	P Value for Comparison Between Groups
Preoperative				
SBP, mean (SD), mm Hg	155 (22)	156 (21)	139 (16)	.001
DBP, mean (SD), mm Hg	91 (15)	92 (14)	82 (12)	.006
Preoperative hypertension grade, No. (%) ^a				
0	35 (19.2)	26 (15.7)	9 (56.3)	.001
1	69 (37.9)	65 (39.2)	4 (25.0)	
2	54 (29.7)	51 (30.7)	3 (18.8)	
3	24 (13.2)	24 (14.5)	0	
No. of antihypertensives/d, median (range) ^b	3 (0 to 7)	3 (0 to 7)	2 (0 to 6)	.06
DDD (n = 173) ^b				
Median (range)	4.3 (0.0 to 22.3)	4.3 (0.0 to 22.3)	4.0 (0.0 to 12.0)	.43
No. of pills/d (n = 173) ^b				
Median (range)	4 (0 to 9)	4 (0 to 9)	2 (0 to 9)	.02
Postoperative				
SBP, mean (SD), mm Hg	147 (15)	149 (13)	122 (13)	<.001
DBP, mean (SD), mm Hg	89 (11)	90 (10)	74 (9)	<.001
Preoperative hypertension grade, No. (%) ^a				
0	16 (8.8)	0	16 (100)	<.001
1	117 (64.3)	117 (70.5)	0	
2	41 (22.5)	41 (24.7)	0	
3	8 (4.4)	8 (4.8)	0	
No. of antihypertensives/d, median (range) ^b	2 (0 to 6)	1 (0 to 6)	3.5 (1 to 6)	<.001
DDD (n = 166) ^b				
Median (range)	2.0 (0.0 to 9.8)	1.7 (0.0 to 7.5)	3.4 (1.0 to 9.8)	.01
No. of pills/d (n = 168) ^b				
Median (range)	2 (0 to 9)	1 (0 to 8)	4 (0 to 9)	.001
Preoperative-Postoperative Change				
SBP, mean (SD), mm Hg	9 (22) ^c	8 (23) ^c	16 (17) ^c	.16
DBP, mean (SD), mm Hg	3 (15) ^c	2 (15)	8 (15) ^c	.12
SBP, mean (SD), %	4.3 (14.0) ^c	3.6 (14.1) ^c	10.9 (11.6) ^c	.05
DBP, mean (SD), %	1.5 (15.1) ^c	0.9 (14.7)	8.3 (17.4) ^c	.06
No. of antihypertensives/d ^b				
Median (range)	1 (−3 to 5) ^c	2 (−2 to 5) ^c	0 (−3 to 0)	<.001
Median (range), %	50 (−100 to 100) ^c	50 (−100 to 100) ^c	0 (−100 to 0)	<.001
DDD (n = 160) ^b				
Median (range)	2.0 (−4.3 to 20.7) ^c	2.0 (−4.3 to 20.7) ^c	0.0 (−4.0 to 9.5)	.006
Median (range), %	53 (−400 to 100) ^c	60 (−256 to 100) ^c	0 (−400 to 79)	.001
No. pills/d (n = 163) ^b				
Median (range)	2 (−3 to 6) ^c	2 (−3 to 6) ^c	0 (−3 to 1)	<.001
Median (range), %	50 (−100 to 100) ^c	50 (−100 to 100) ^c	0 (−100 to 50)	<.001

Abbreviations: DBP, diastolic blood pressure; DDD, defined daily dose; ESH, European Society of Hypertension; JNC, Joint National Commission; SBP, systolic blood pressure.

^a Grade 0, SBP less than 140 mm Hg and DBP less than 90 mm Hg; grade 1, SBP 140 to 159 mm Hg and/or DBP 90 to 99 mm Hg; grade 2, SBP 160 to 179 mm Hg and/or DBP 100 to 109 mm Hg; and grade 3, SBP 180 mm Hg or higher or DBP 110 mm Hg or higher.^{20,21}

^b Values not normally distributed given as medians (range).

^c Significant ($P < .05$) preoperative-postoperative change.

stratified as no clear success, this decrease in SBP was 10 mm Hg or more without an increase in the use of antihypertensives. As shown by Ettehad et al,¹⁹ this reduction should be considered clinically relevant because every 10-mm Hg reduction in SBP leads to a risk reduction of 20% in major cardiovascular events, 17% in coronary heart disease, 27% in stroke, 28% in heart failure, and 13% in all-cause mortality. Combining these results shows that, after adrenalectomy for PA, 90% of patients had any form of decrease in BP and/or number of antihypertensive medications and, in a minimum of 73% of patients, we considered this decrease as clinically significant.

Numerous studies targeted clinical success by describing proportions of patients with clinical cure and/or clinical improvement with a large heterogeneity in outcome criteria. In our study, 27.1% of patients showed clinical cure, which is lower

to my for PA, 90% of patients had any form of decrease in BP and/or number of antihypertensive medications and, in a minimum of 73% of patients, we considered this decrease as clinically significant.

Table 5. Outcome of Surgery Stratified Based on Magnitude of Change in SBP

Variable	No. (%)			
	Total Cohort (n = 435)	Clinical Success		
		Cure (n = 118)	Clear Improvement (n = 135)	No Clear Success (n = 182)
Increase in SBP	76 (17.5)	12 (10.2) ^a	8 (5.9) ^a	56 (30.8)
Decrease in SBP, mm Hg				
0-9	71 (16.3)	21 (17.8)	13 (9.6)	37 (20.3)
10-19	87 (20.0)	22 (18.6)	36 (26.7)	29 (15.9)
20-29	84 (19.3)	29 (24.6)	35 (25.9)	20 (11.0)
30-39	51 (11.7)	19 (16.1)	19 (14.1)	13 (7.1)
40-49	24 (5.5)	7 (5.9)	12 (8.9)	5 (2.7)
≥50	28 (6.4)	8 (6.8)	12 (8.9)	8 (4.4)
Increase in No. of antihypertensives ^b	14 (3.2)	0	0	14 (7.7)

Abbreviation: SBP, systolic blood pressure.

^a Despite an increase in SBP after surgery, these patients were normotensive (SBP <140 and diastolic blood pressure <90 mm Hg) during no antihypertensive therapy or a lower number of antihypertensives.

^b Owing to an increase in the number of antihypertensive medications, a decrease in the SBP could be due to medication instead of surgery. Therefore, these patients were excluded from analysis.

than the 42%, 50%, and 52% cure rates presented in reviews and meta-analyses.¹³⁻¹⁵ However, most studies in these reviews and meta-analyses were small, single-center, and included patients over a wide range of years or even decades. Furthermore, because most studies focused on describing proportions of patients with clinical cure and/or improvement and potential prognostic factors, they presented no or limited data on the magnitude of decrease in BP and number of antihypertensive medications, making data regarding this subject scarce.

However, recently, results from the Primary Aldosteronism Surgery Outcome (PASO) investigators were published.¹⁸ They presented clinical outcomes of adrenalectomy in a large, worldwide cohort of patients who underwent surgery for PA between 1994 and 2015. Although their primary goal was to establish consensus criteria for clinical and biochemical outcomes and describe prognostic factors for each outcome, they also displayed some data regarding the magnitude of decrease in BP and number of antihypertensive medications within each outcome definition.¹⁸

The PASO investigators showed complete cure in 37% of patients, which is lower compared with the above-mentioned reviews and meta-analyses,¹³⁻¹⁵ but higher compared with our study. Although preoperative BP measurements were comparable between studies, the PASO investigators also showed a larger decrease in mean (SD) SBP and DBP compared with our results (SBP: 22 [22] vs 17 [21] mm Hg, $P < .001$; and DBP: 11 [14] vs 7 [14] mm Hg, $P < .001$). These differences may be attributable to dissimilar baseline characteristics. In accordance with earlier performed studies, the PASO investigators identified younger age, female sex, and lower body mass index as indicators of a favorable clinical outcome.^{18,23,24}

Although age and distribution of sex were comparable between studies, body mass index (calculated as weight in kilograms divided by height in meters squared) was lower within the PASO study cohort: 27.8 (5.2) vs 29.7 (6.0) ($P < .001$).¹⁸ Therefore, difference in body mass index possibly could be of influence. Because our study is representative for current clinical practice diagnostic modalities, such as adrenal venous sampling or a confirmatory test, were performed based on a cen-

ter's preference and availability rather than routinely in all patients. Therefore, patient selection also could be a influencing factor. Another possible factor could be the substantial number of patients with relatively short follow-up after surgery within our cohort. In our cohort, however, the period of follow-up was not a significant factor influencing the proportions of patients with cure, clear improvement, and no clear success ($P = .28$). Further comparison with the PASO study was not possible owing to different definitions of clear improvement and no clear success.

In contrast to earlier studies, we present clinical outcomes, including data on magnitude of BP decrease, after adrenalectomy for PA in a large cohort including only patients who underwent surgery within recent years. We chose to limit the sample to minimize the potential risk on bias because of possible improvements in diagnosis, workup, and treatment of PA owing to innovation of guidelines, diagnostics modalities, and surgical techniques over time. Furthermore, as indicated by Namekawa et al,²⁵ an increase in the prevalence of obesity and diabetes over the past few decades potentially leads to a decrease in favorable clinical outcomes. Likewise, owing to the worldwide increase of hypertension within the past decades, patients are less likely to achieve clinical cure of hypertension that is due to background or essential hypertension, which is not PA related.¹⁶ Therefore, including patients over a wide range of years or even decades could lead to overestimation of surgical outcomes compared with results in current clinical practice. Another strength of this study is the worldwide, multicenter design that makes us believe that our results may be representative of the Western world.

Limitations

Similar to almost all other studies regarding PA, the need for a retrospective design, mostly because of the low prevalence of PA, is one of the weaknesses of our study, especially since this design made it impossible to use standardized procedures for BP measurements, such as performing out-of-office measurements in all patients. Also, the substantial number of patients with a relatively short follow-up is a potential shortcoming of this study. However, because excluding these patients with shorter follow-up could introduce selection bias and

the duration of follow-up had no significant influence on our primary outcomes, we chose to not exclude patients based on follow-up duration.

Conclusions

Decreased BP and reduced need for antihypertensive medications, in addition to biochemical cure, appear to be clinically

relevant benefits of adrenalectomy in patients with PA. Although this study shows complete clinical cure in only approximately one-quarter to one-third of the included patients, most patients became normotensive while receiving lower or equal use of antihypertensive medications. Moreover, a large proportion of the patients with persistent hypertension after surgery may benefit from adrenalectomy given the observed clinically relevant and significant reduction of BP and antihypertensive use.

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